Day 128 | Unit 09 | Teacher plan

### Display, Describe, and Summarize Data

# Find Mean Absolute Deviation

Standards: 6.SP.B.5c

Learning target: I can calculate and interpret mean absolute deviation.

enVision Topic 8: Display, Describe, and Summarize Data | Spectrum companion Chapter 7

## 0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

## 5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Find the mean. Calculate each value’s distance from the mean, add those distances, then divide by the number of observations. MAD describes a typical distance from the mean in the original units.

## 17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

## 42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

## Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

## Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references:

Day 128 | Unit 09 | Lesson

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# Find Mean Absolute Deviation

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## 01 / Learn

Find the mean. Calculate each value’s distance from the mean, add those distances, then divide by the number of observations. MAD describes a typical distance from the mean in the original units.

## Worked example 1

Reading times in minutes are 6, 7, 7, 8, 8, 9. Calculate the mean and MAD. Show all absolute deviations.

1. Find the mean.

2. Calculate each value’s distance from the mean, add those distances, then divide by the number of observations.

3. Distances: 3/2, 1/2, 1/2, 1/2, 1/2, 3/2

4. sum 5

5. divide by 6 = 5/6.

6. State the result in the form requested: Mean 15/2 min; MAD 5/6 min.

Answer: Mean 15/2 min; MAD 5/6 min

## Worked example 2

Six students recorded reading times (minutes): 4, 7, 7, 10, 10, 13. Part A: enter the mean. Part B: enter the MAD and interpret it in minutes.

1. Find the mean.

2. Mean = 51/6 = 17/2.

3. Absolute deviations: 9/2, 3/2, 3/2, 3/2, 3/2, 9/2.

4. Their sum divided by 6 is 5/2.

5. This is the average distance from the mean.

Answer: Mean 17/2 min; MAD 5/2 min

# Day 128 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

### 01 / Constructed response

Reading times in minutes are 5, 6, 6, 7, 7, 8. Calculate the mean and MAD. Show all absolute deviations.

### 02 / Constructed response

Reading times in minutes are 7, 8, 8, 9, 9, 10. Calculate the mean and MAD. Show all absolute deviations.

### 03 / Constructed response

Reading times in minutes are 2, 3, 3, 4, 4, 5. Calculate the mean and MAD. Show all absolute deviations.

### 04 / Constructed response

Reading times in minutes are 3, 6, 6, 9, 9, 12. Calculate the mean and MAD. Show all absolute deviations.

### 05 / Constructed response

Reading times in minutes are 5, 9, 9, 13, 13, 17. Calculate the mean and MAD. Show all absolute deviations.

### 06 / Constructed response

Reading times in minutes are 8, 14, 14, 20, 20, 26. Calculate the mean and MAD. Show all absolute deviations.

## 07 / Constructed response

Reading times in minutes are 2, 3, 3, 4, 5, 14. Calculate the mean and MAD. Show all absolute deviations. Interpret MAD in a sentence.

## 08 / Numeric entry and interpretation

Six students recorded reading times (minutes): 2, 8, 8, 14, 14, 20. Part A: enter the mean. Part B: enter the MAD and interpret it in minutes.

## 09 / Numeric entry and interpretation

Six students recorded reading times (minutes): 4, 6, 6, 8, 8, 10. Part A: enter the mean. Part B: enter the MAD and interpret it in minutes.

## 10 / Constructed response

Reading times in minutes are 5, 6, 6, 7, 8, 17. Calculate the mean and MAD. Show all absolute deviations. Interpret MAD in a sentence.

Day 128 | Unit 09 | Answer key

### Display, Describe, and Summarize Data

# Find Mean Absolute Deviation

Standards: 6.SP.B.5c

Learning target: I can calculate and interpret mean absolute deviation.

### 1. Mean 13/2 min; MAD 5/6 min

Distances: 3/2, 1/2, 1/2, 1/2, 1/2, 3/2; sum 5; divide by 6 = 5/6.

6.SP.B.5c | Constructed response

### 2. Mean 17/2 min; MAD 5/6 min

Distances: 3/2, 1/2, 1/2, 1/2, 1/2, 3/2; sum 5; divide by 6 = 5/6.

6.SP.B.5c | Constructed response

### 3. Mean 7/2 min; MAD 5/6 min

Distances: 3/2, 1/2, 1/2, 1/2, 1/2, 3/2; sum 5; divide by 6 = 5/6.

6.SP.B.5c | Constructed response

### 4. Mean 15/2 min; MAD 5/2 min

Distances: 9/2, 3/2, 3/2, 3/2, 3/2, 9/2; sum 15; divide by 6 = 5/2.

6.SP.B.5c | Constructed response

### 5. Mean 11 min; MAD 10/3 min

Distances: 6, 2, 2, 2, 2, 6; sum 20; divide by 6 = 10/3.

6.SP.B.5c | Constructed response

### 6. Mean 17 min; MAD 5 min

Distances: 9, 3, 3, 3, 3, 9; sum 30; divide by 6 = 5.

6.SP.B.5c | Constructed response

### 7. Mean 31/6 min; MAD 53/18 min; a typical distance from the mean is 53/18 min

Distances: 19/6, 13/6, 13/6, 7/6, 1/6, 53/6; sum 53/3; divide by 6 = 53/18.

6.SP.B.5c | Constructed response

## 8. Mean 11 min; MAD 5 min

Mean = 66/6 = 11. Absolute deviations: 9, 3, 3, 3, 3, 9. Their sum divided by 6 is 5. This is the average distance from the mean.

6.SP.B.5c | Numeric entry and interpretation

## 9. Mean 7 min; MAD 5/3 min

Mean = 42/6 = 7. Absolute deviations: 3, 1, 1, 1, 1, 3. Their sum divided by 6 is 5/3. This is the average distance from the mean.

6.SP.B.5c | Numeric entry and interpretation

## 10. Mean 49/6 min; MAD 53/18 min; a typical distance from the mean is 53/18 min

Distances: 19/6, 13/6, 13/6, 7/6, 1/6, 53/6; sum 53/3; divide by 6 = 53/18.

6.SP.B.5c | Constructed response